

Session 2 Coding Assignment

Build a Qubit State Explorer using Qiskit

Aditya Singhal

B.Tech Artificial Intelligence and Machine Learning

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1 Objective

The objective of this assignment is to understand the effect of different quantum gates on the state of a single qubit. The program was developed using Python and Qiskit to allow users to select a quantum gate, apply it to a qubit, visualize the resulting state using the Bloch Sphere, and observe the corresponding measurement probabilities.

2 Introduction

Quantum computing is based on quantum bits (qubits), which can exist in a superposition of states unlike classical bits. Quantum gates are mathematical operations that manipulate qubits and change their quantum state.

In this assignment, the following quantum gates were explored:

- X Gate
- Y Gate
- Z Gate
- Hadamard (H) Gate
- RY Rotation Gate

The Qiskit library was used to simulate these operations and visualize the resulting qubit state.

3 Software Used

The following software and libraries were used during implementation.

- Google Colab
- Python
- Qiskit
- NumPy
- Matplotlib

4 Methodology

The implementation follows the following steps:

1. Create a quantum circuit with one qubit.
2. Display a menu for gate selection.
3. Accept the user's choice.
4. Ask for a rotation angle if the RY gate is selected.
5. Apply the selected quantum gate.
6. Generate the statevector.
7. Calculate measurement probabilities.
8. Display the quantum circuit.
9. Visualize the final state using the Bloch Sphere.

This workflow makes the program interactive and easy to understand for beginners.

5 Experimental Results

5.1 X Gate

The X gate flips the qubit from $|0\rangle$ to $|1\rangle$. Therefore, the probability of measuring $|1\rangle$ becomes 1.

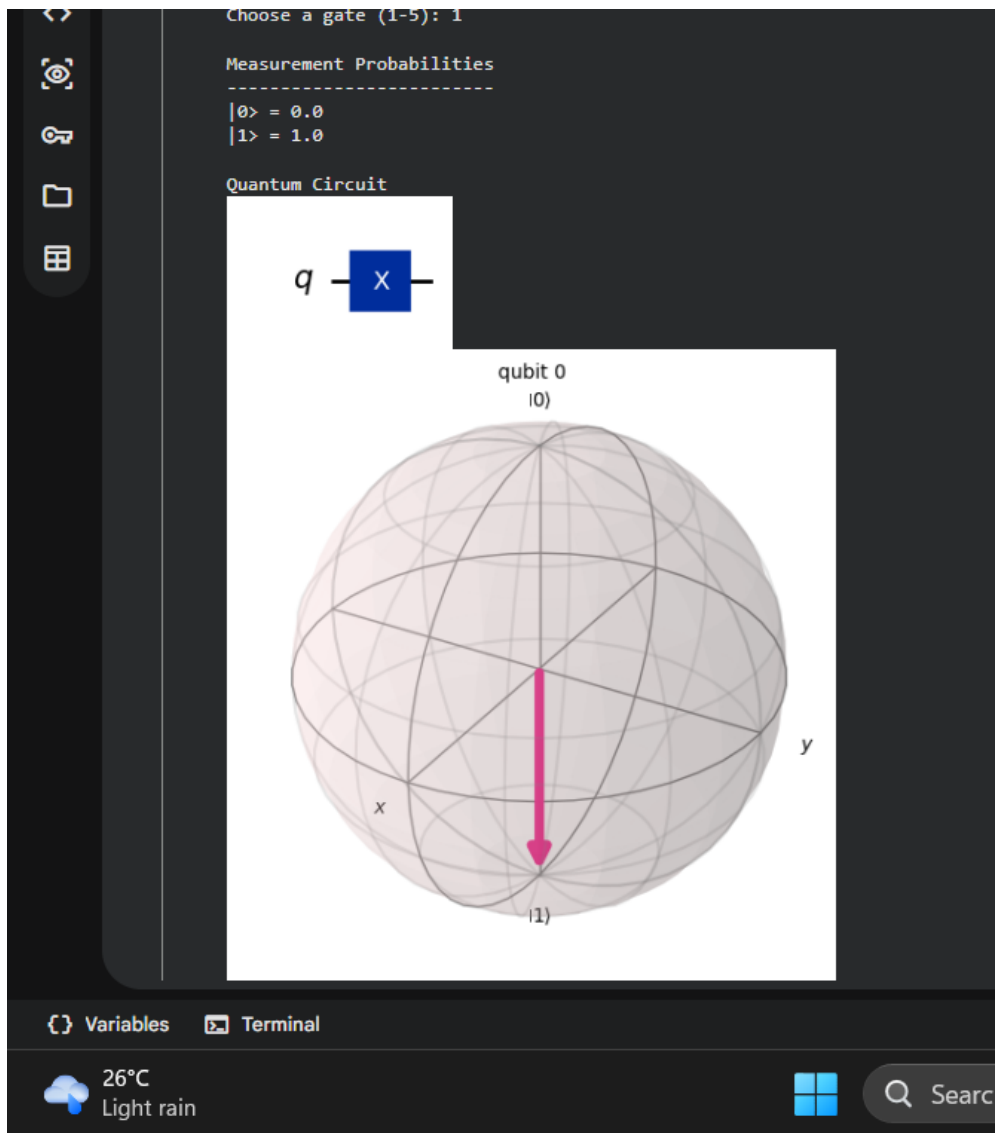


Figure 1: Output after applying the X Gate.

5.2 Y Gate

The Y gate performs a rotation around the Y-axis along with a phase change. Starting from $|0\rangle$, the measurement outcome becomes $|1\rangle$.

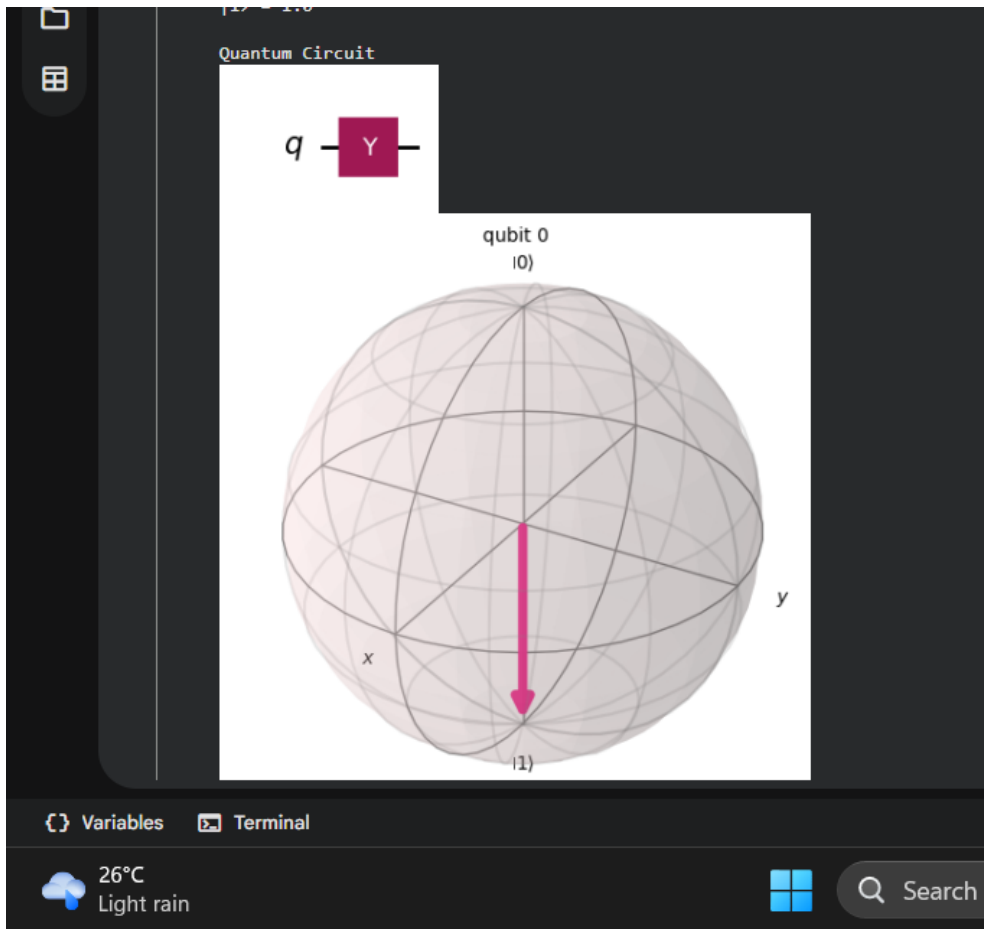


Figure 2: Output after applying the Y Gate.

5.3 Z Gate

The Z gate changes only the phase of the qubit while keeping the measurement probabilities unchanged for the initial state.

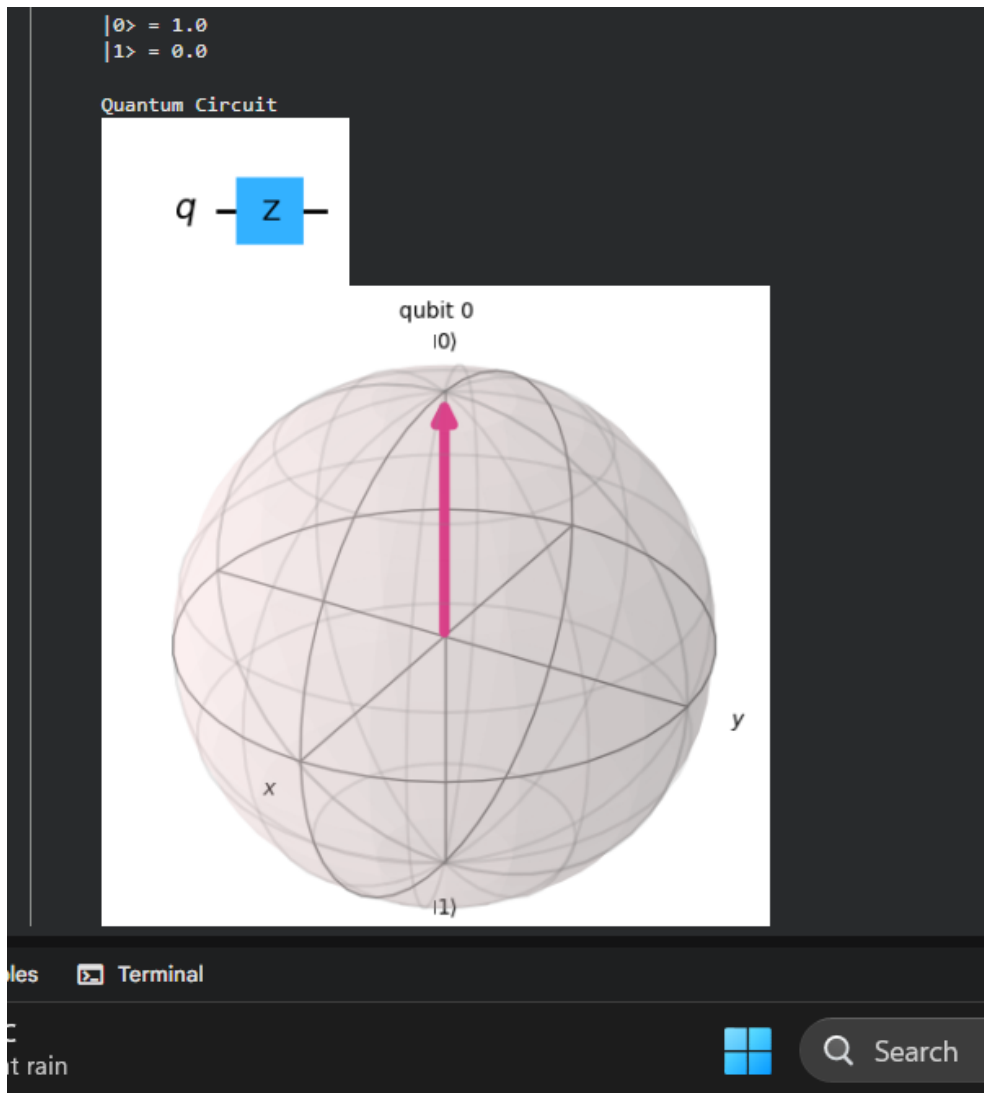


Figure 3: Output after applying the Z Gate.

5.4 Hadamard Gate

The Hadamard gate creates an equal superposition of the $|0\rangle$ and $|1\rangle$ states, resulting in approximately equal measurement probabilities.

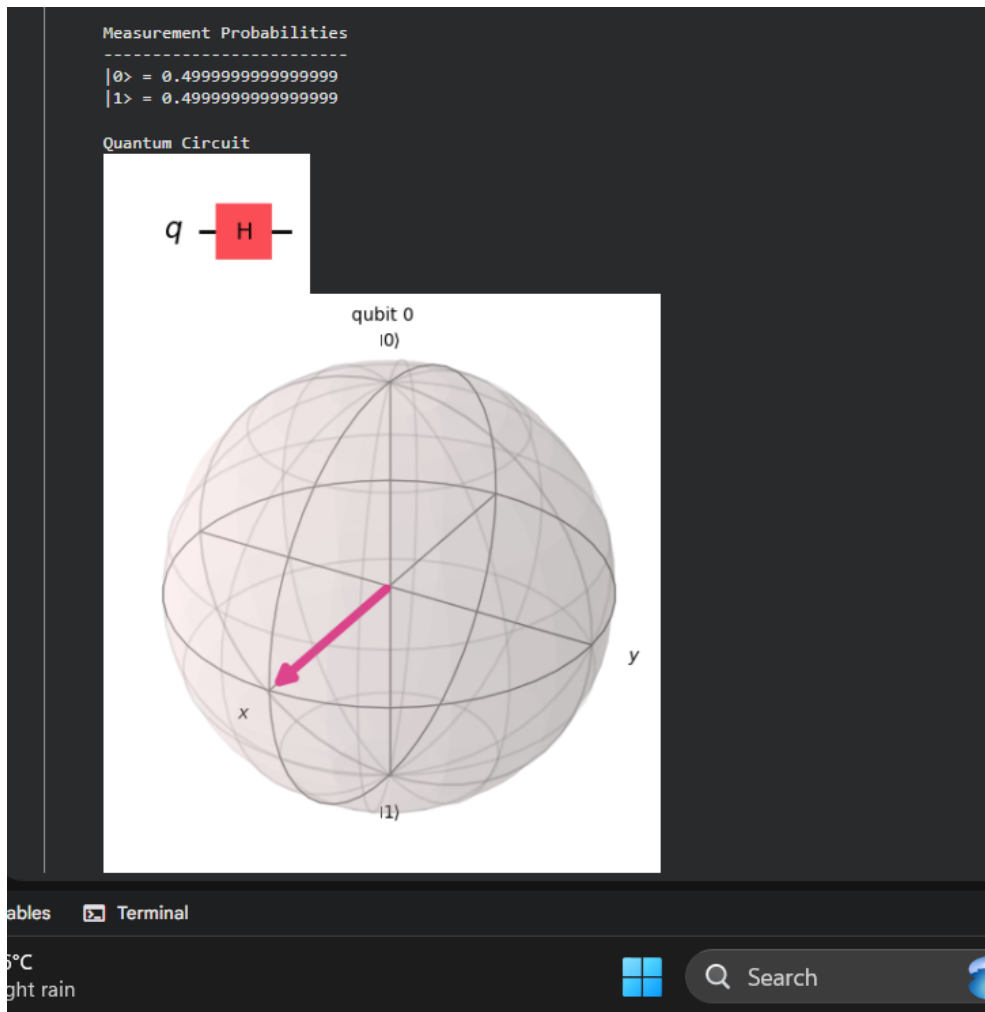


Figure 4: Output after applying the Hadamard Gate.

5.5 RY Gate

The RY gate rotates the qubit around the Y-axis by a user-defined angle. The measurement probabilities vary depending on the chosen rotation angle.

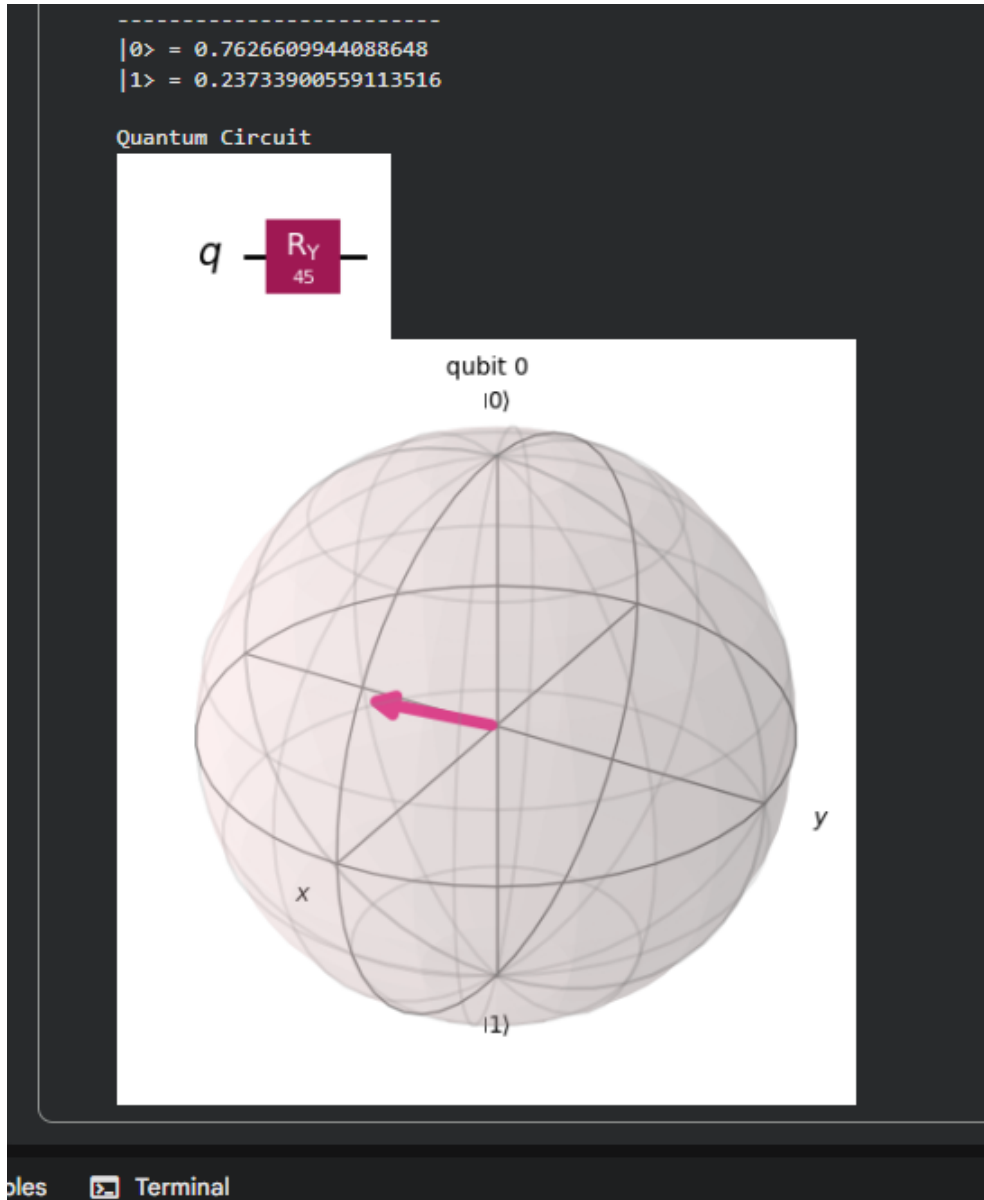


Figure 5: Output after applying the R_Y Gate.

6 Observations

The following observations were made during the execution of the program.

- The X gate flips the qubit completely from $|0\rangle$ to $|1\rangle$.
- The Y gate also changes the measurement outcome to $|1\rangle$ while introducing a phase difference.
- The Z gate modifies only the phase of the qubit and therefore keeps the measurement probabilities unchanged.

- The Hadamard gate creates an equal superposition, giving probabilities close to 0.5 for both basis states.
- The RY gate allows continuous rotation of the qubit and produces different probabilities depending on the rotation angle entered by the user.
- The Bloch Sphere provides a clear visualization of how the state vector moves after each quantum operation.

7 Conclusion

This assignment helped in understanding the behavior of a single qubit under different quantum gate operations. Using Qiskit, it became easier to visualize how each gate transforms the quantum state through both the Bloch Sphere and measurement probabilities.

The project strengthened the basic concepts of quantum computing such as state representation, superposition, phase change, and quantum rotations. It also provided practical experience with the Qiskit framework and quantum circuit simulation.

GitHub Repository

The complete source code, report, and output screenshots have been uploaded to the GitHub repository as required.

Repository Link

<https://github.com/YourUsername/Qubit-State-Explorer>

Replace the above link with your actual GitHub repository URL before submission.